

I have included two different project proposals. The first one would be a real-world project where I set up my own business, developing a pet water foundation including programming the microcontrollers (Arduino, Raspberry Pi Pico) responsible for the automated foundation functions, standing up and securing a company website to host information about the device, and potentially an app to directly control the device. This is a passion project for me which would tackle both the Network Engineering and Cybersecurity portion of my degree and would push my knowledge of programming to a new level. This project proposal would be considered high risk and high reward as there's a lot of unknown variables and the timeline for this project may have to be adjusted several times. If you allow it, I'd pick this project first.

[Project Proposal 1 – Link here.](#)

The second project proposal would be what you listed in the project ideas for capstone projects, "Research an attack method (malspam with attached malicious document) and design a detection method for it (IDS rules, firewall rules, network analysis)" as this sounds very interesting and would definitely push my cybersecurity knowledge to the next level. This would be my second choice as a project.

[Project Proposal 2 – Link here.](#)